

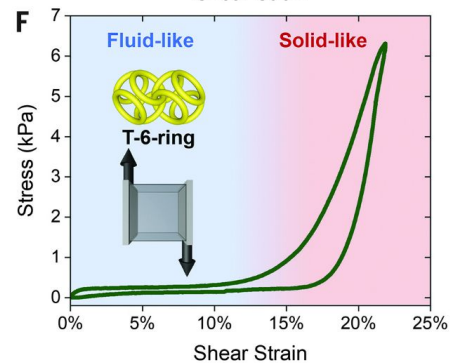
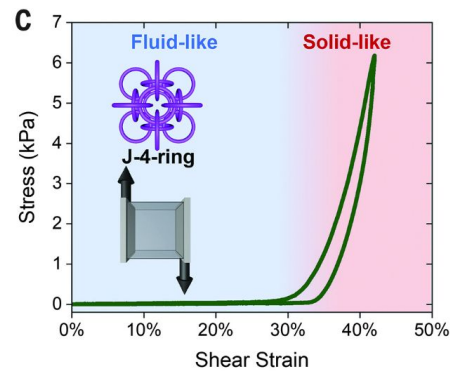
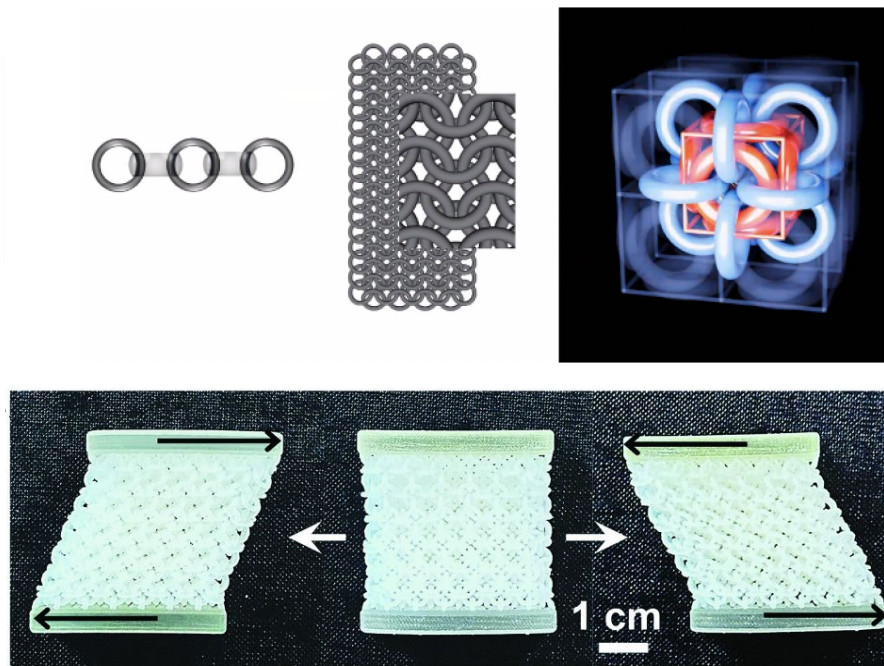
Experimental Platform for Investigating Dynamics in Polycatenated Architected Materials (PAMs)

Kensuke Shimojo (Carl F. Braun SURF Fellow)
Mentors: Alice Xiong, Prof. Chiara Daraio

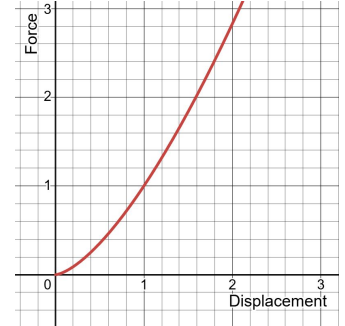
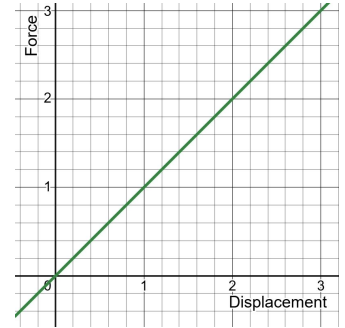
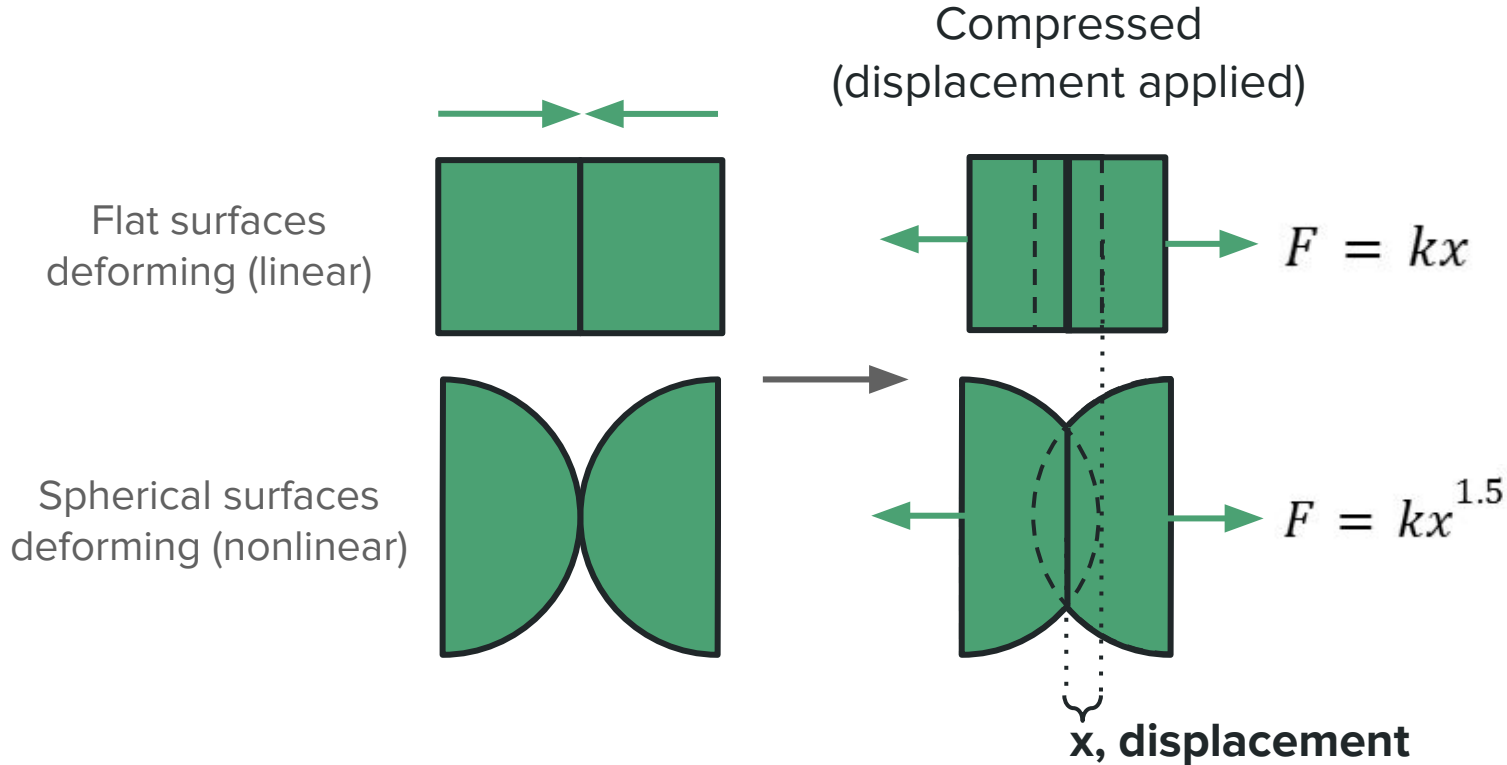
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 - Pulse velocity vs. excitation amplitude
- **Future Directions**
 - Junction setup
 - Other modes of actuation

Polycatenated Architected Materials (PAMs)

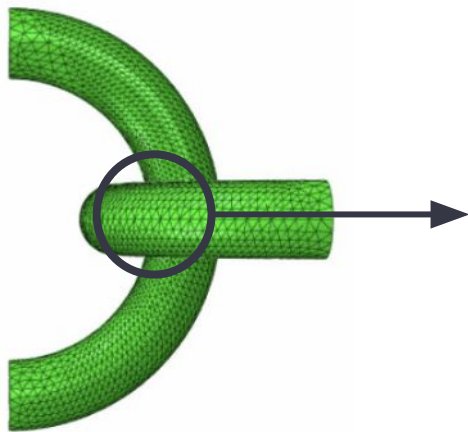


Theory of PAMs: Nonlinear Force vs. Displacement

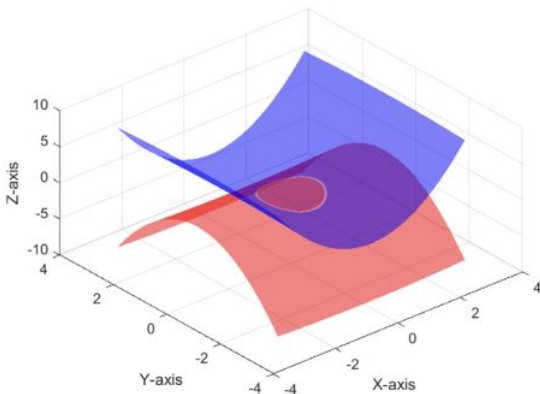


Theory of PAMs: Nonlinear Springs at Linkages

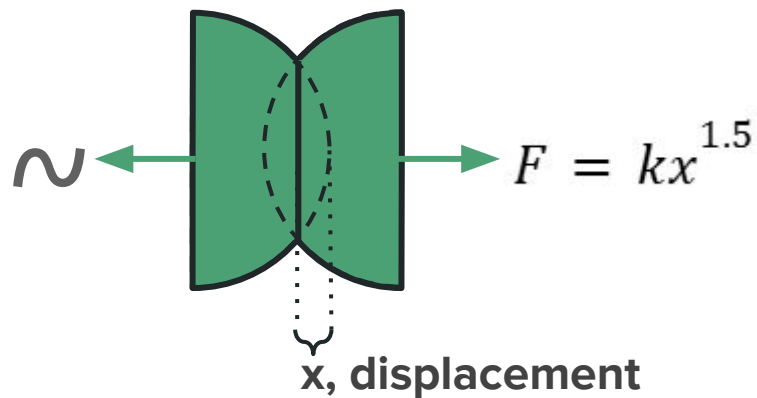
Circular Ring Contacts



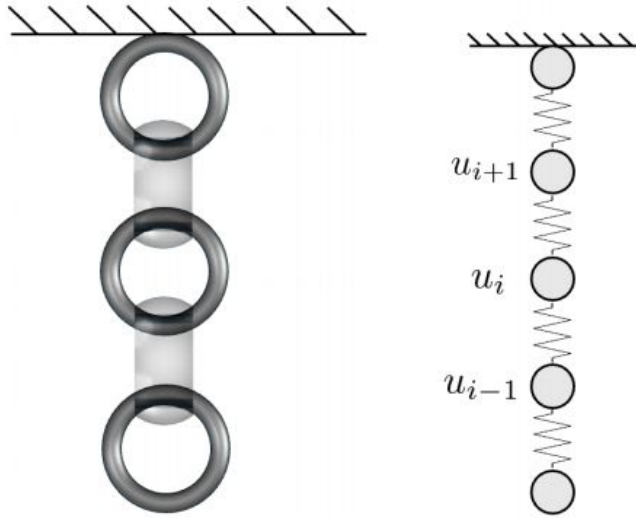
2 Saddles Contacting and Deforming



2 Spheres Contacting and Deforming



Theory of PAMs: PDE Setup



Treat a linkage between PAM units as a nonlinear spring:

$$F = kx^n$$

where $n=1.5$ for our case

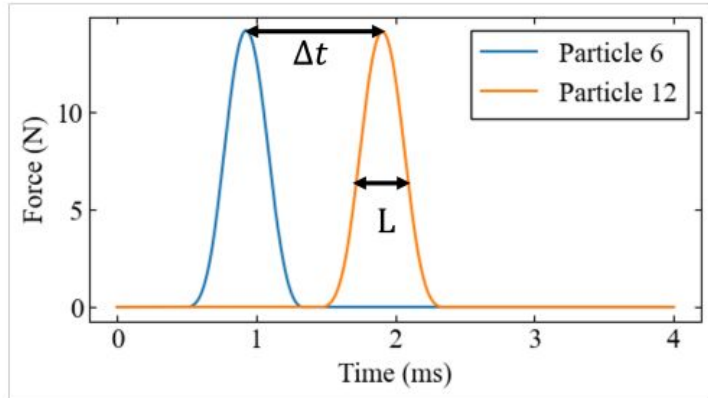
Model the multiple mass-spring system using a system of partial differential equations (PDEs) based on:

$$F_{net} = m\ddot{u}_i = k(u_{i+1} - u_i + \delta_T)^n - k(u_i - u_{i-1} + \delta_T)^n$$

where: u_i is the displacement of the i -th mass from its equilibrium position, $\ddot{u}_i$ is acceleration of the i -th mass, and δ_T is the displacement due to pretension

Here, $(u_i - u_{i-1} + \delta_T) = x$, the displacement of the spring

Theory of PAMs: Solitary Wave Solutions



Seek for solutions where sharp impulse is applied at boundary → Solution is a **solitary wave** (pulse) that travels at velocity $\mathbf{V}_s$:

$$V_s = \sqrt{\frac{2}{n+1}} D \left(\frac{k^{1/n}}{m} \right)^{1/2} \left(F_m^{\frac{n-1}{2n}} \right)$$

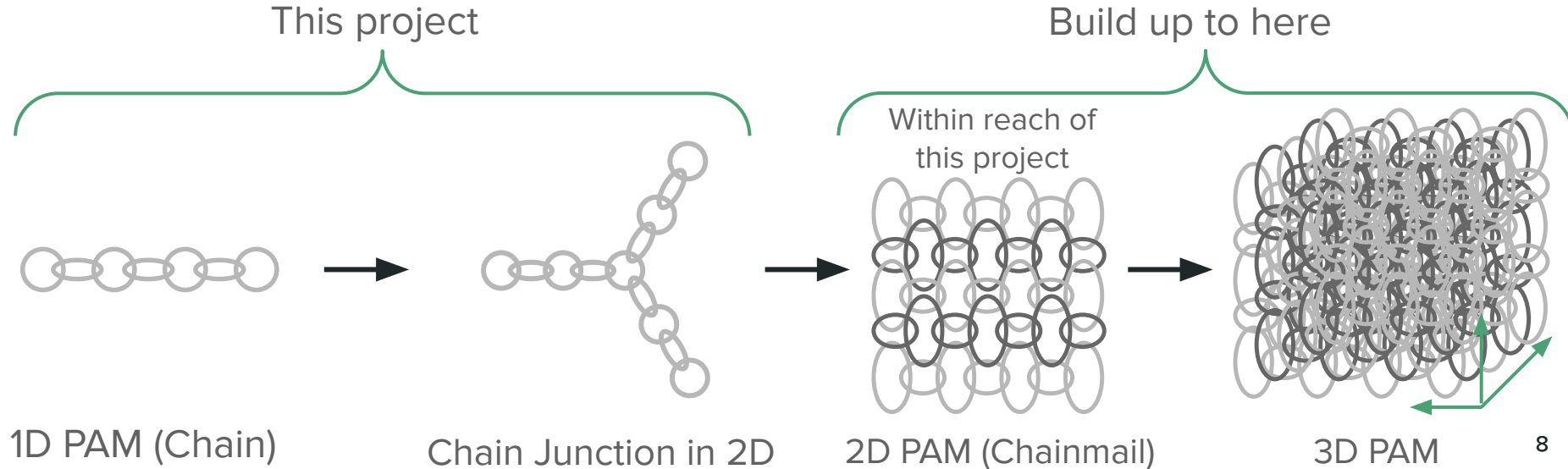
where: $\mathbf{F}_m$ is the maximum force exerted within the PAM units and D is a constant coefficient

$\mathbf{V}_s$ is a nonlinear function of the **maximum force**, which can be controlled by **pretension** or **input amplitude**

Goal of SURF

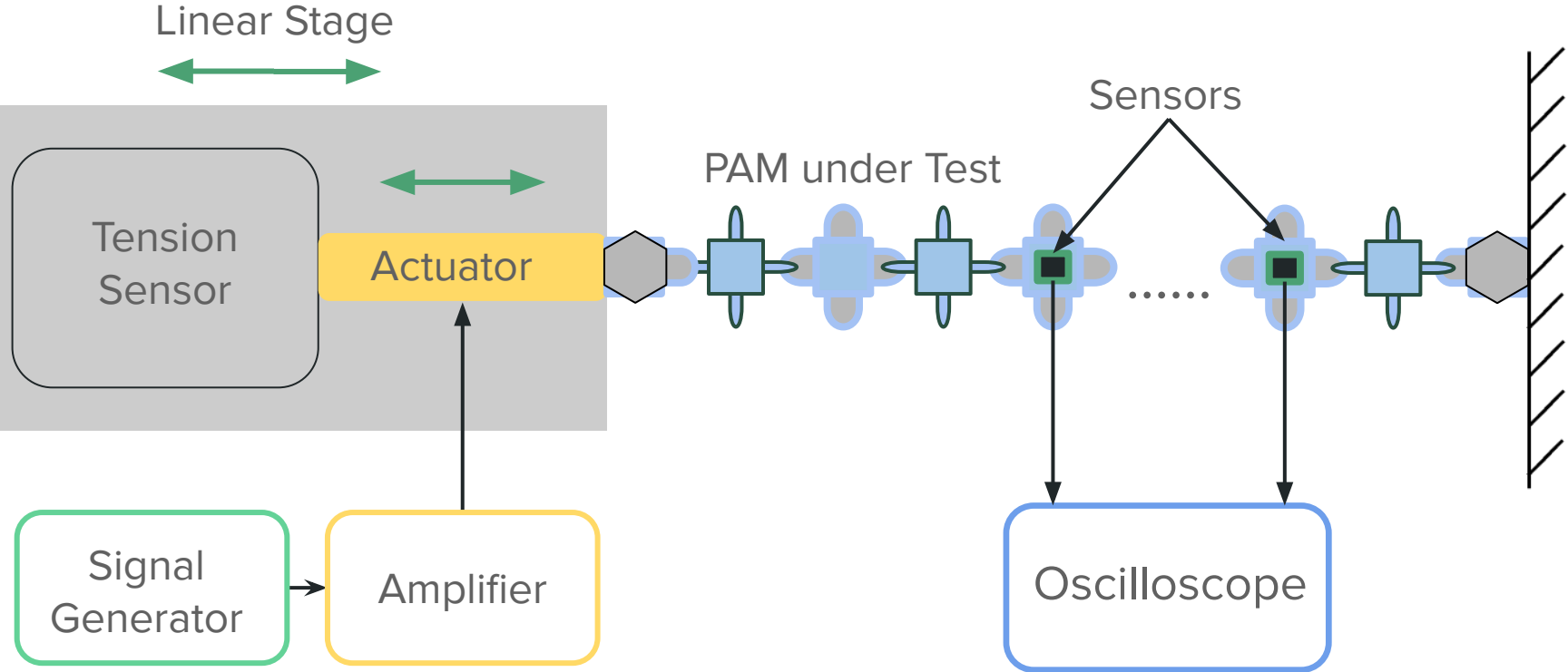
To build an experimental platform that can:

- Measure wave dynamics in PAM structures under various conditions
- Compare and contrast with theoretical model of PAMs

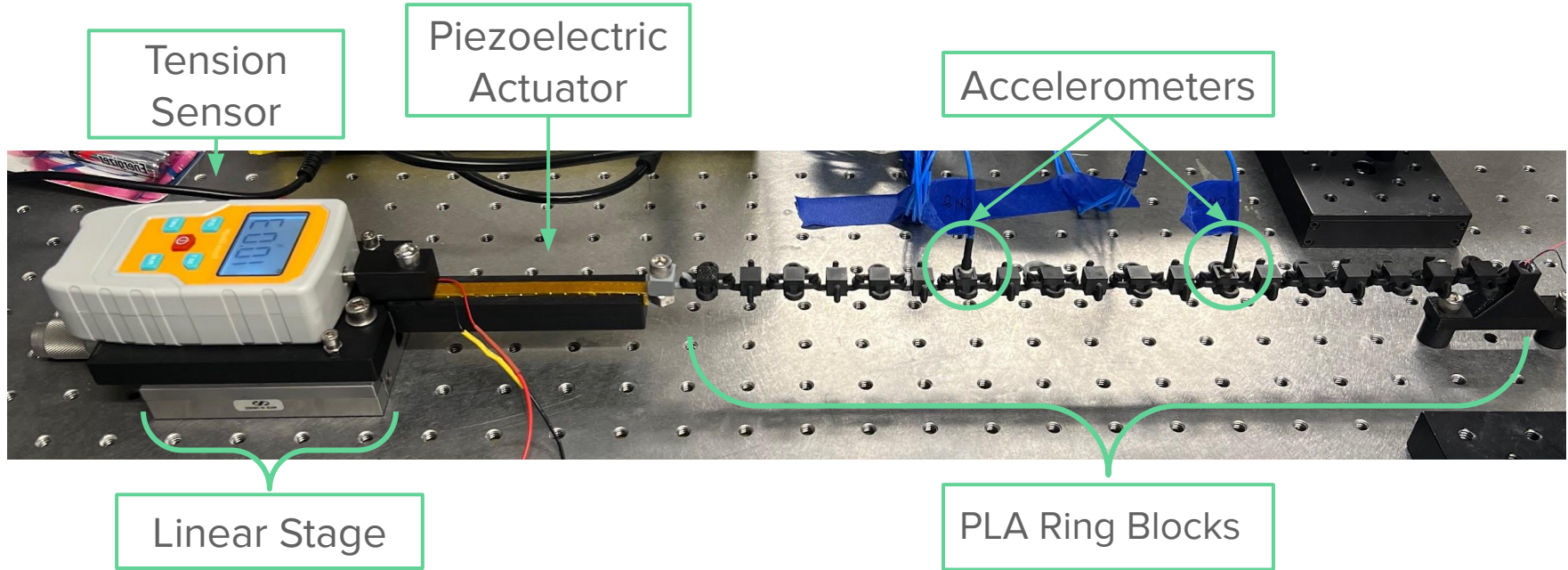


General Idea of the Setup

Experimental Setup Diagram

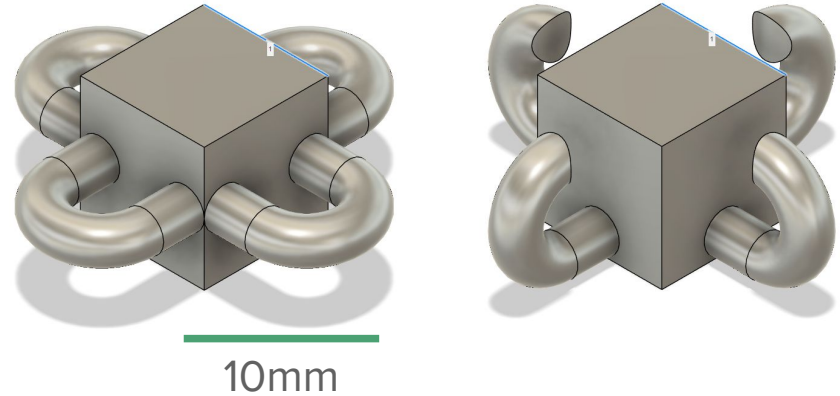


Actual Setup



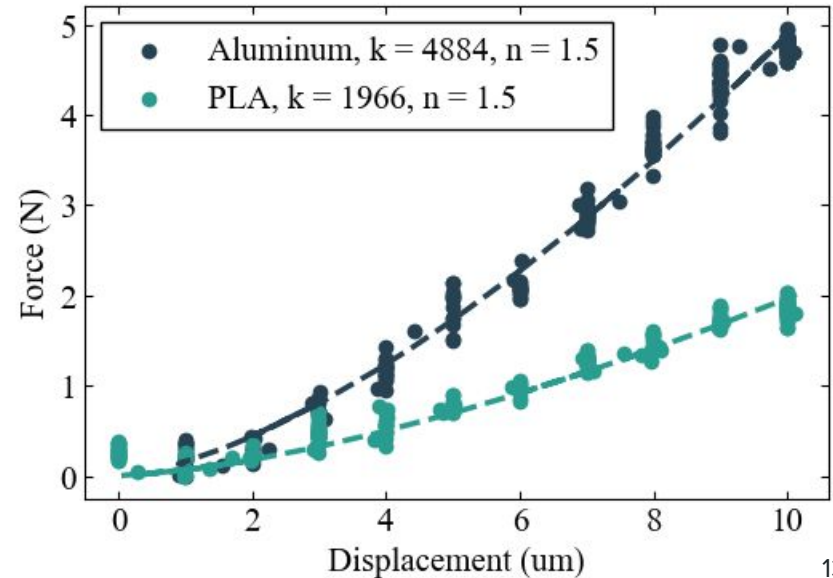
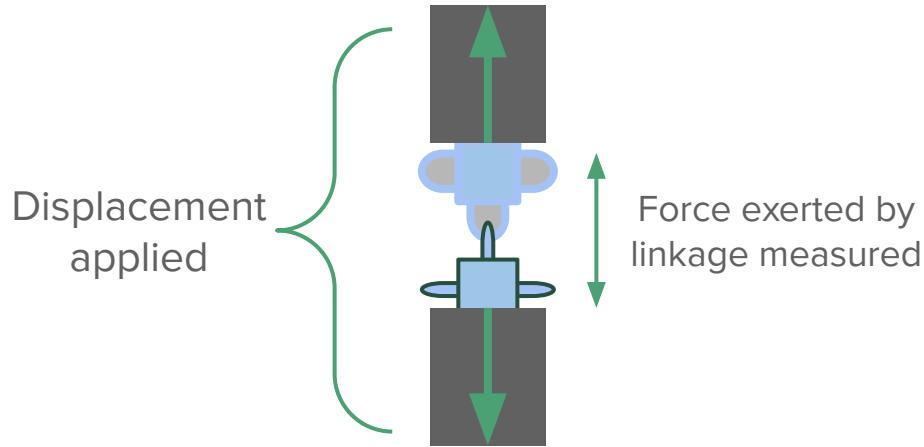
PAM Unit Cell: Ring Blocks

- Size
 - Big enough for material to not yield
 - Small enough for mass production and higher # of linkages
- Geometry
 - Hook shape: greater ease of manufacturing and setup
 - Follows same power law in linkage

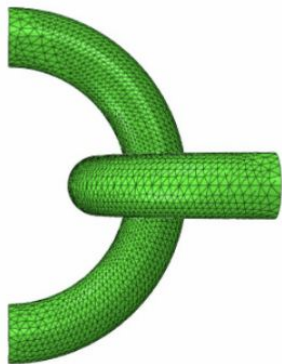


PAM Unit Cell: Testing

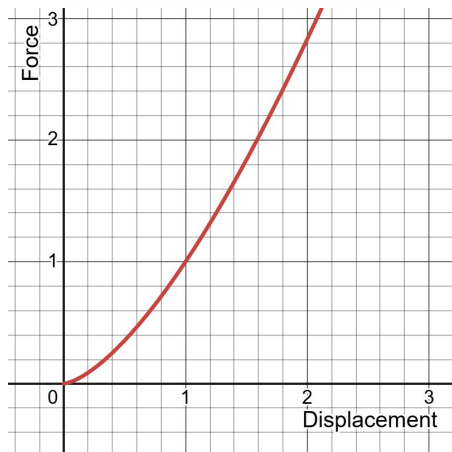
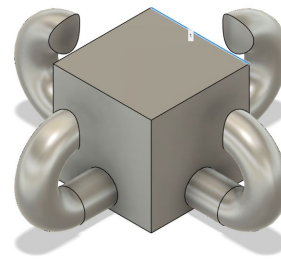
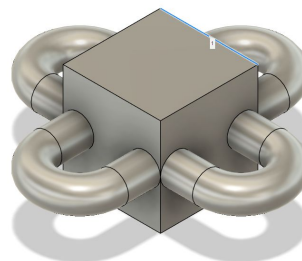
- Tensile test on 2 linked blocks (1 ring contact)
- PLA link shows $F = kx^{1.5}$ relationship: nonlinear spring as desired from linkage



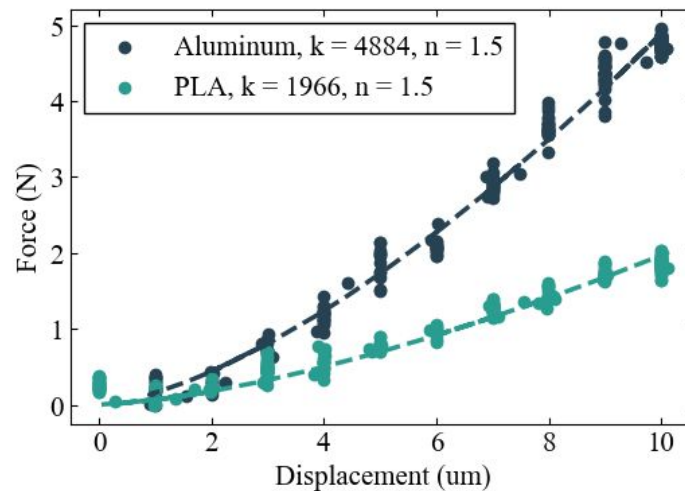
PAM Unit Cell: Ring Blocks $\sim$ Rings



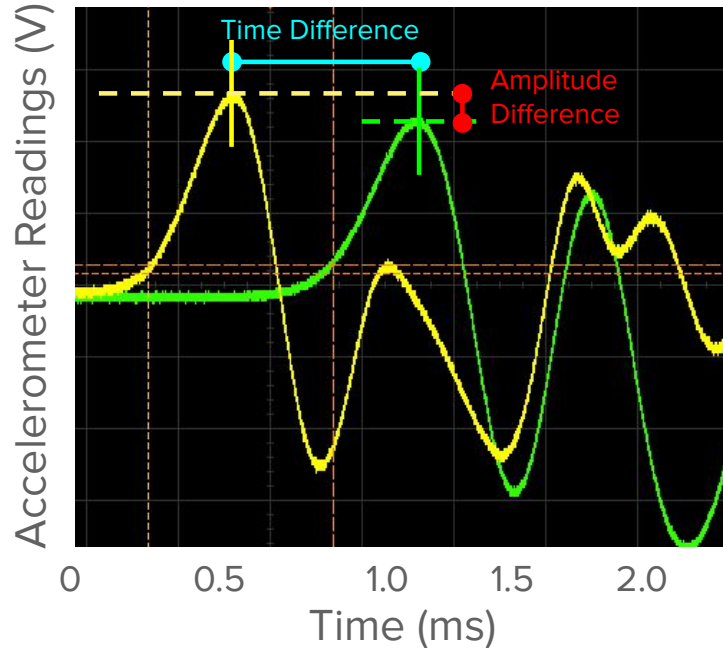
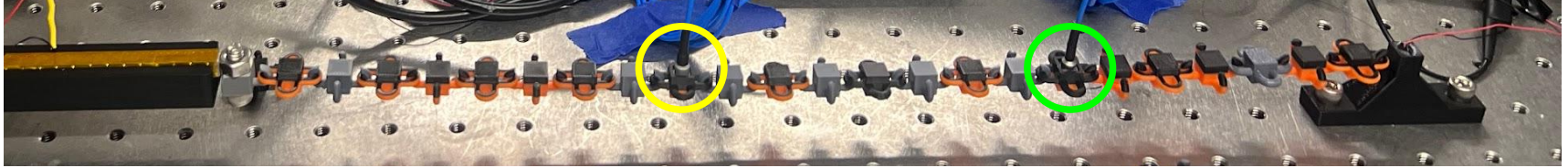
$\sim$



$\sim$



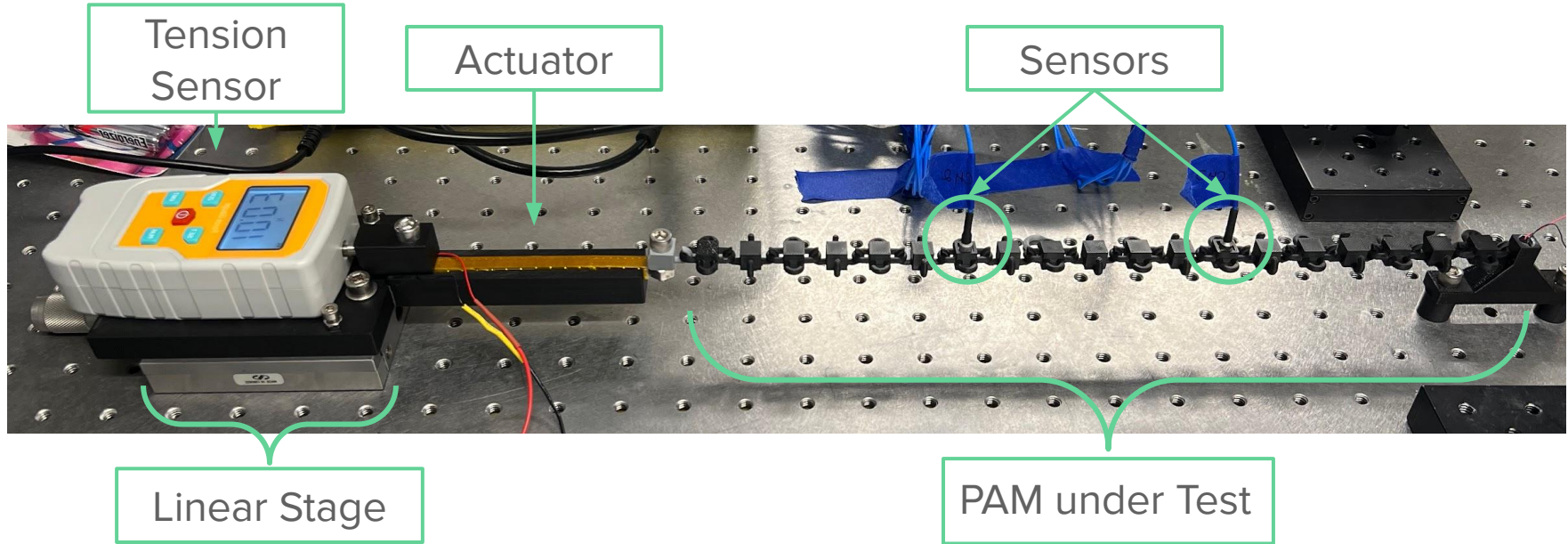
Collecting Data from Accelerometer



From acceleration peaks from 2 sensors, we can get:

- Time difference $\rightarrow$ distance/time = average **velocity** of pulse
- **Amplitude** of pulse
- Amplitude drop $\rightarrow$ energy dissipated per link

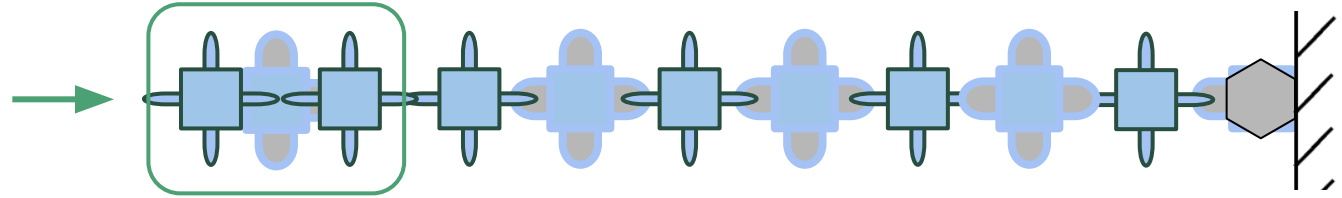
Experimental Setup



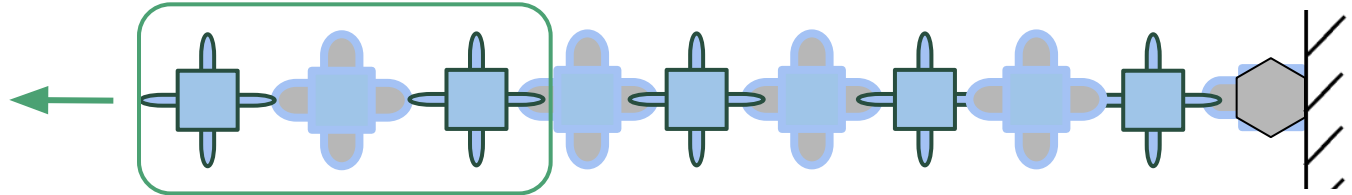
Velocity vs. Pretension Experiment

Hypothesis and Setup

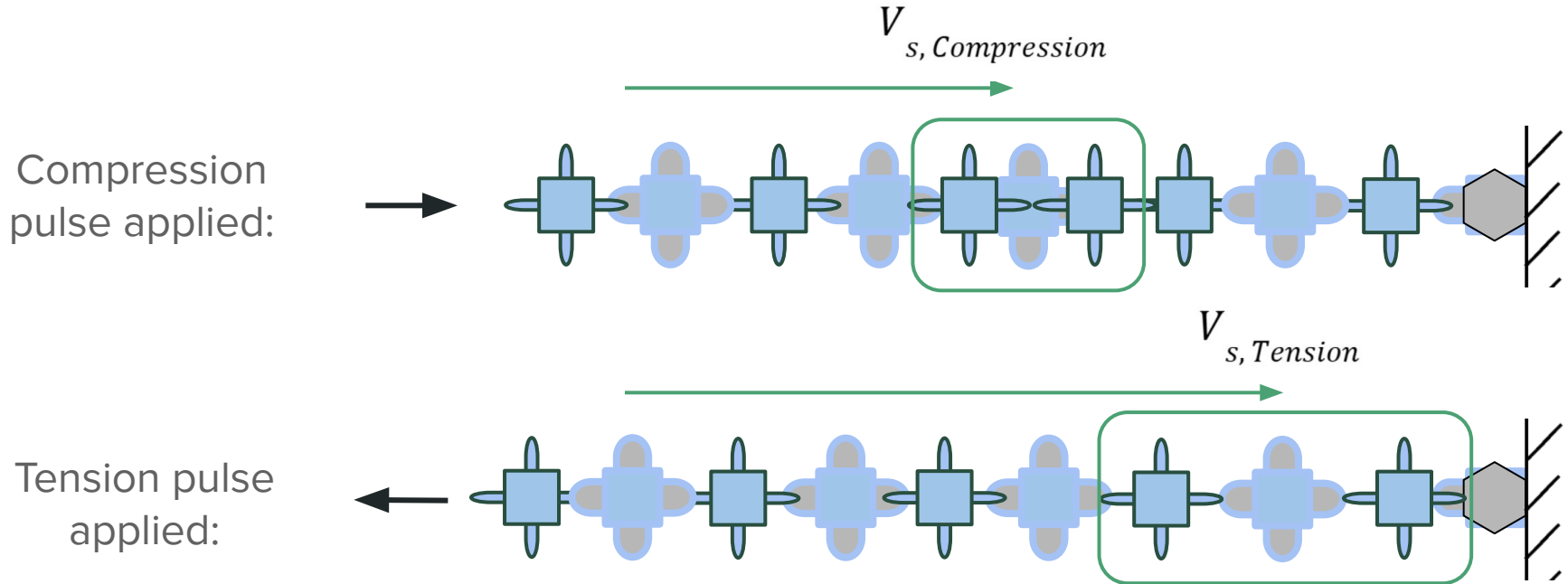
Compression
pulse applied:



Tension pulse
applied:



Hypothesis and Setup

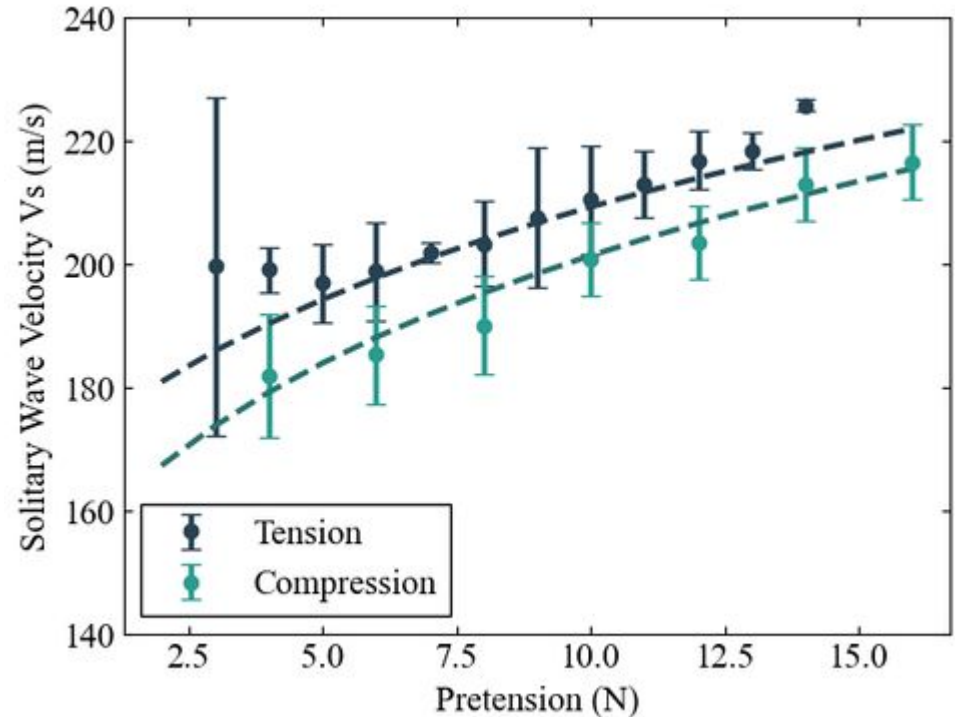


$$V_s = \sqrt{\frac{2}{n+1}} D \left(\frac{k^{1/n}}{m} \right)^{1/2} \left(F_m^{\frac{n-1}{2n}} \right)$$

$$V_{s, Tension} > V_{s, Compression}$$

1D Chain: Solitary Wave Velocity vs Pretension

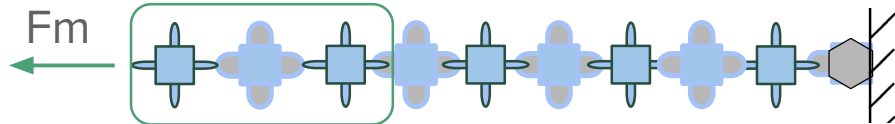
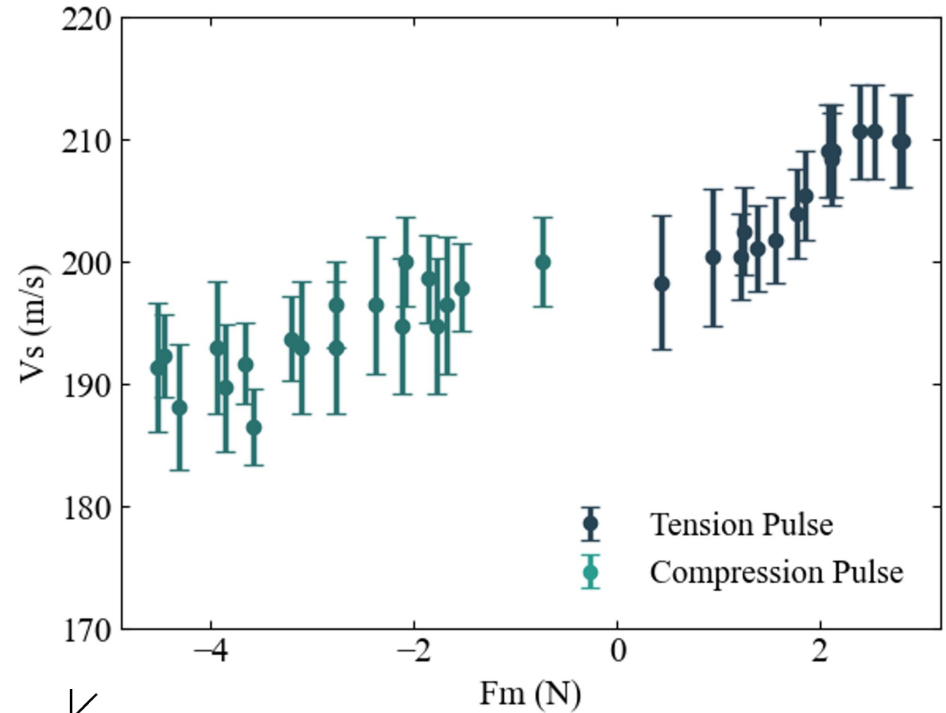
- Fix input amplitude, vary pretension
- Small impulses sent parallel (tension) and anti-parallel (compression) with existing pretension
- Tension pulse is slightly faster than compression pulse



Velocity vs. Amplitude Experiment

1D Chain: Solitary Wave Velocity vs Amplitude

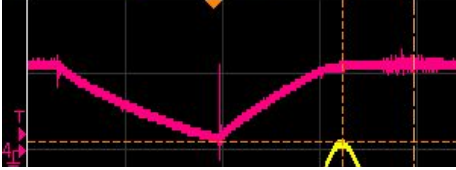
- Fix pretension, vary input amplitude
- F_m is the peak pulse amplitude at the first sensor site
- Tension pulse faster than compression pulse



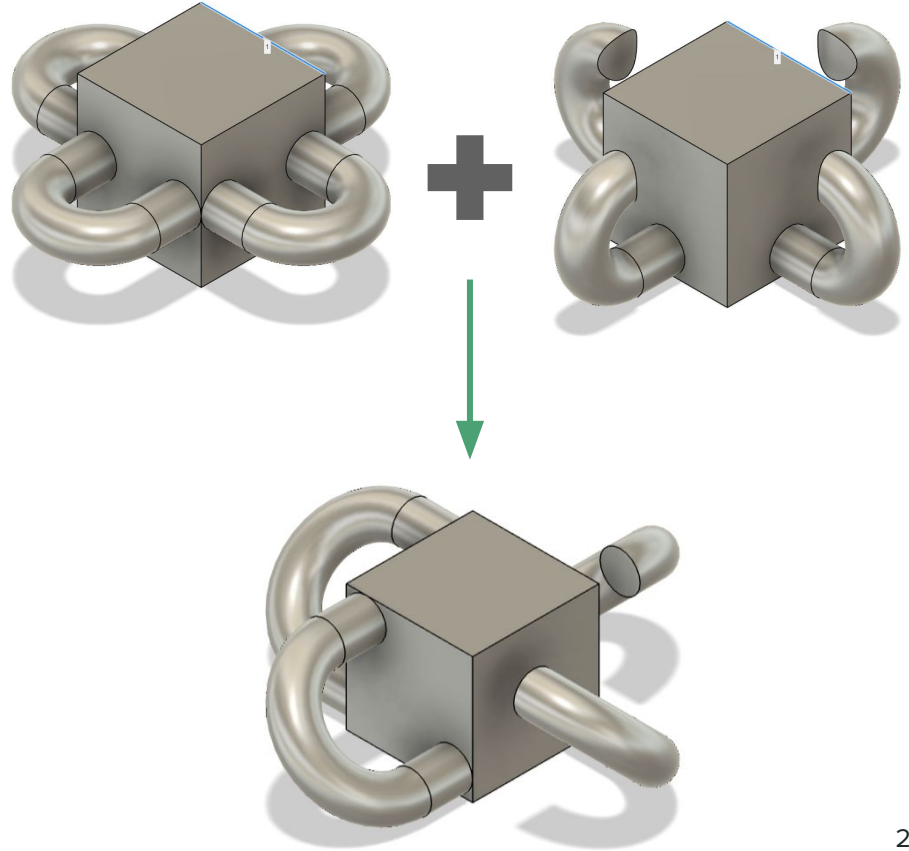
Next Steps

Next Steps: Improvements

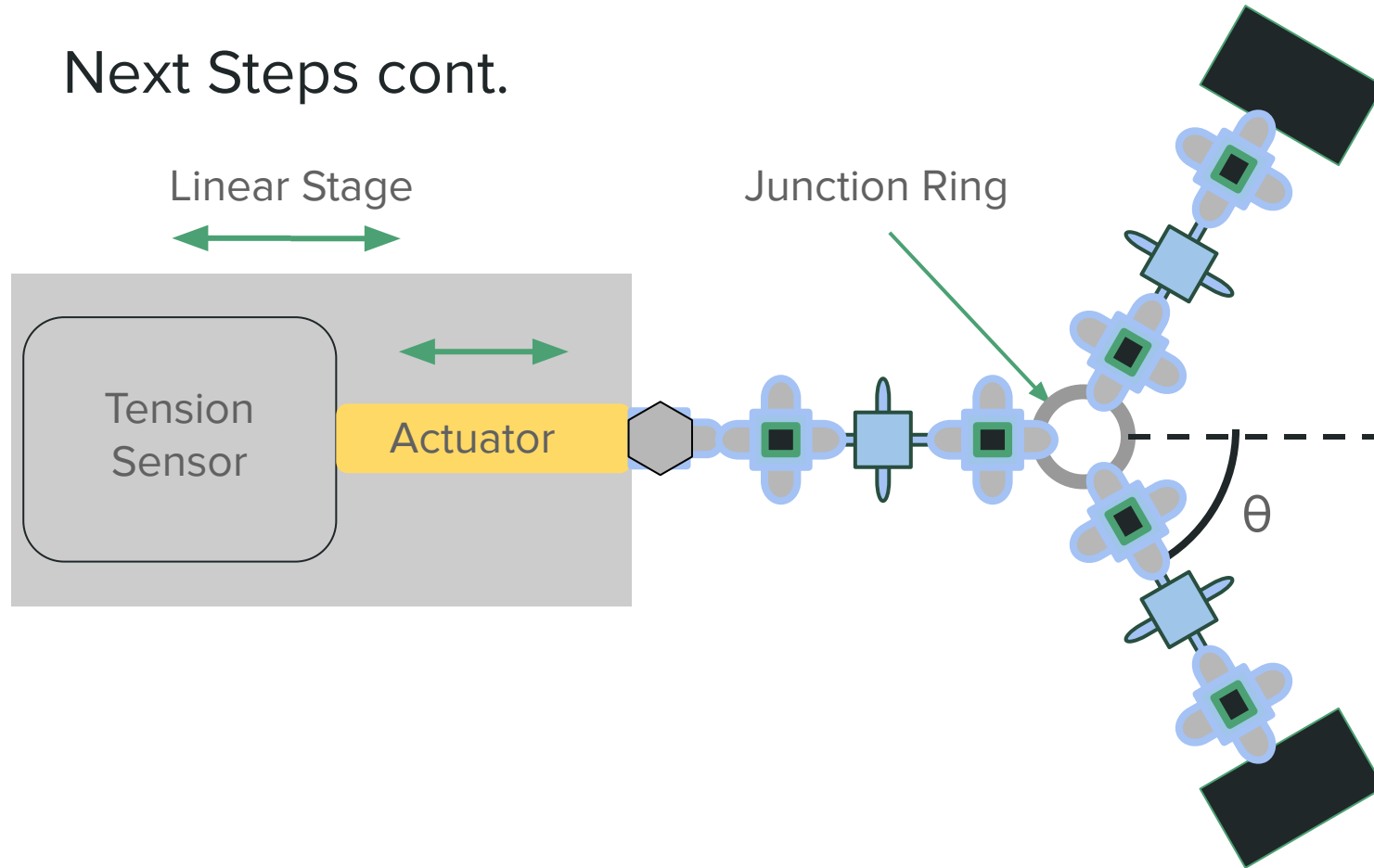
Actuator
Voltage
(Displacement)



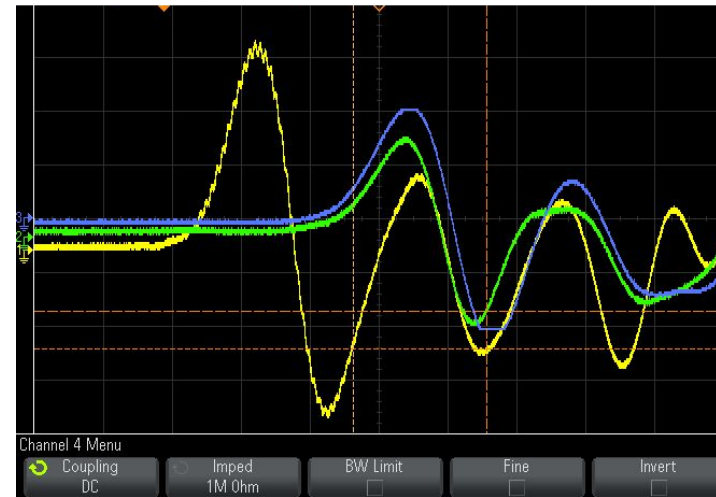
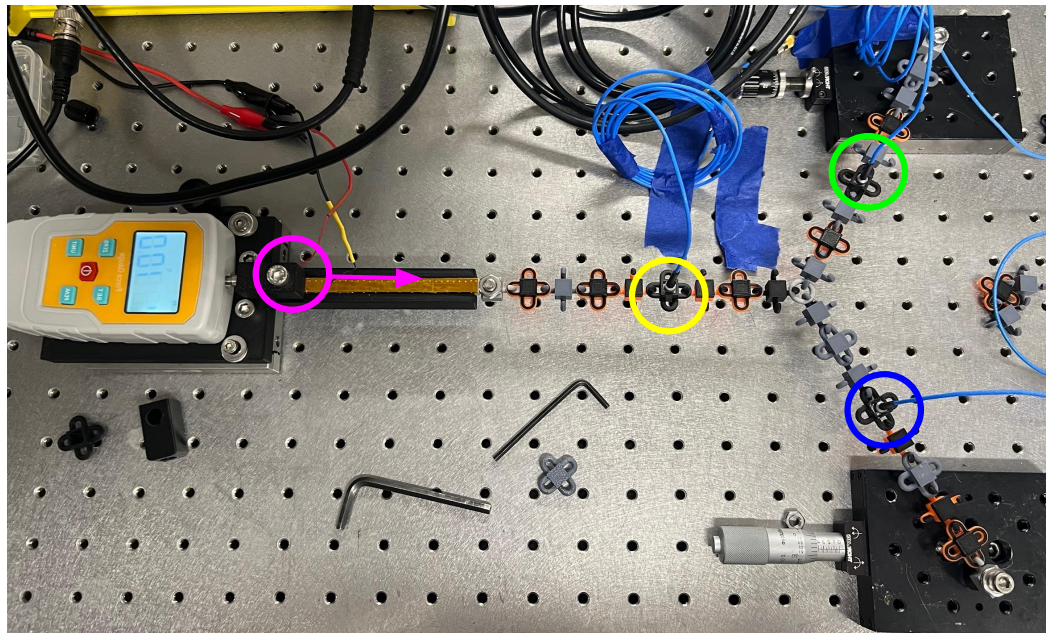
Input
Voltage
(Desired)



Next Steps cont.



Junction Setup



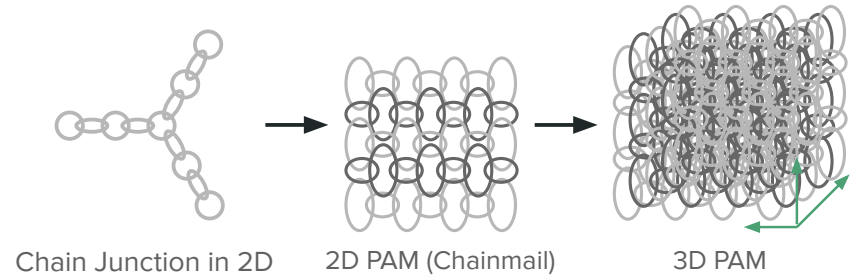
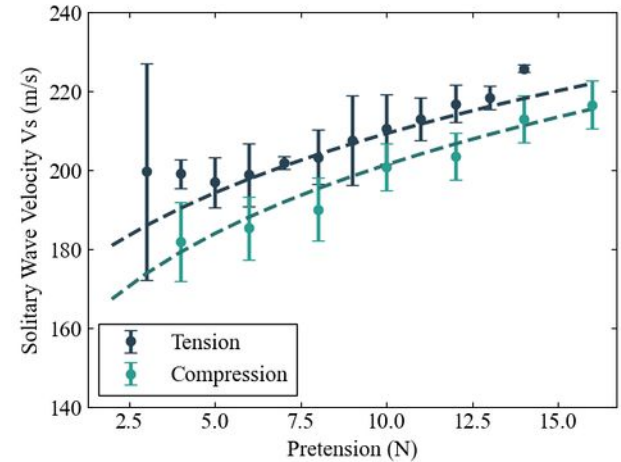
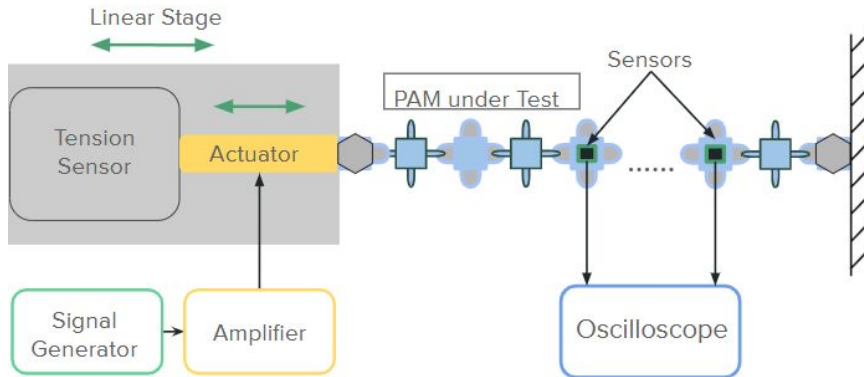
Next Steps Cont.

- Different PAM linkages
- Mix and mash of different material units
- Different modes of actuation and pulses
- Test other waveform inputs (sinusoid, Gaussian, sech^2 , etc.)



Conclusions

- Setup successfully captured interesting data trend, which is consistent with theory- validation
- Experimental setup can be easily modified:
- Serves as foundation for experimental investigations into PAMs
 - Insight into physics of 3D PAMs → usability as tunable material



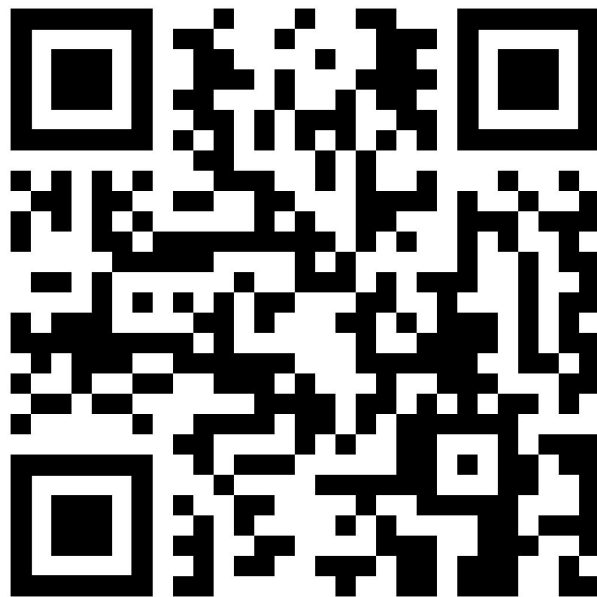
Acknowledgements

Thank you to my mentor Alice, Prof. Daraio, and the chainmail group for your mentorship and guidance, and the Carl F. Braun Trust and SFP for funding this project!

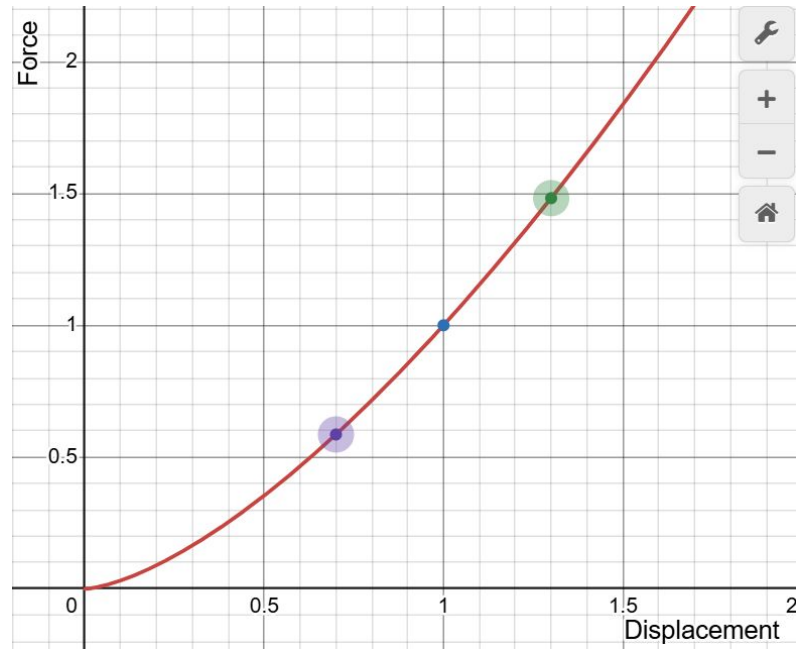


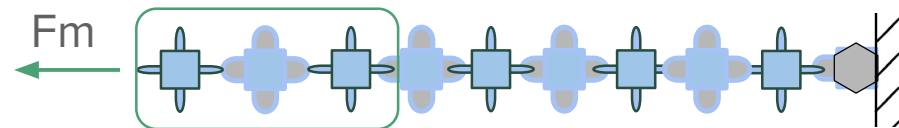
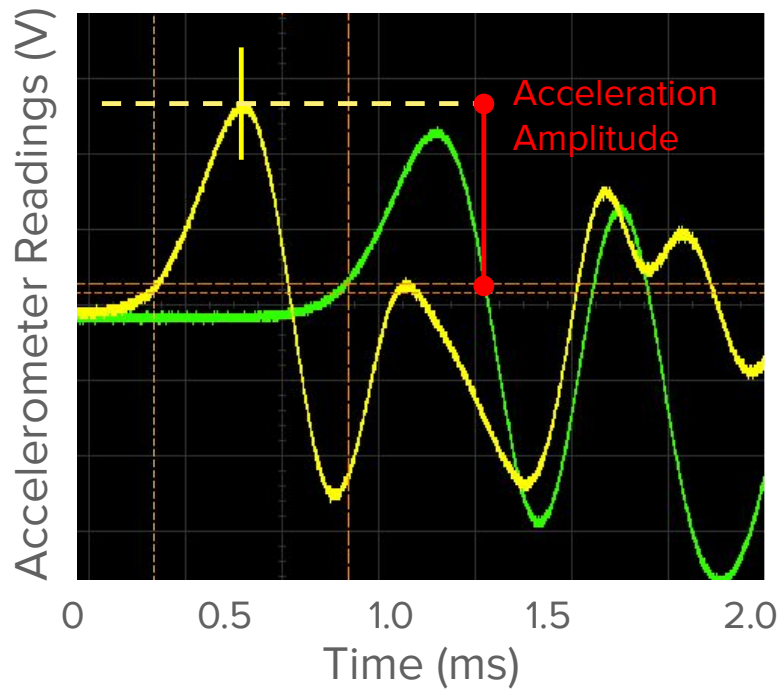
Feedback Form

Name: Kensuke Shimojo

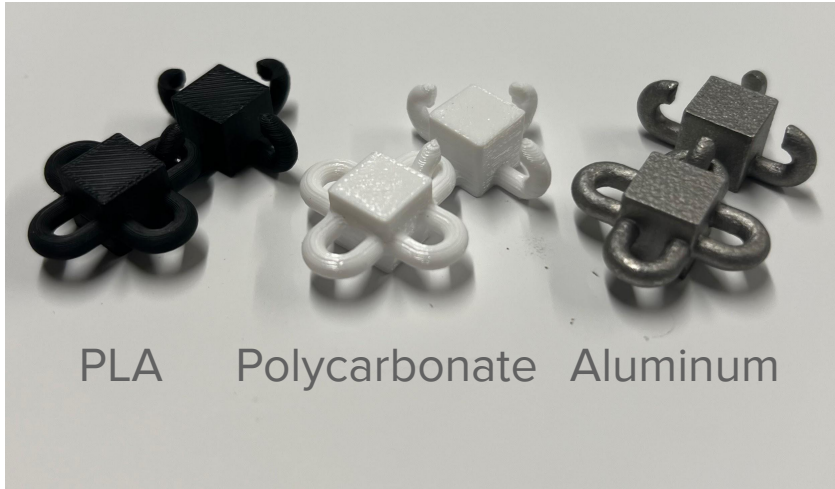


Thank you for listening!





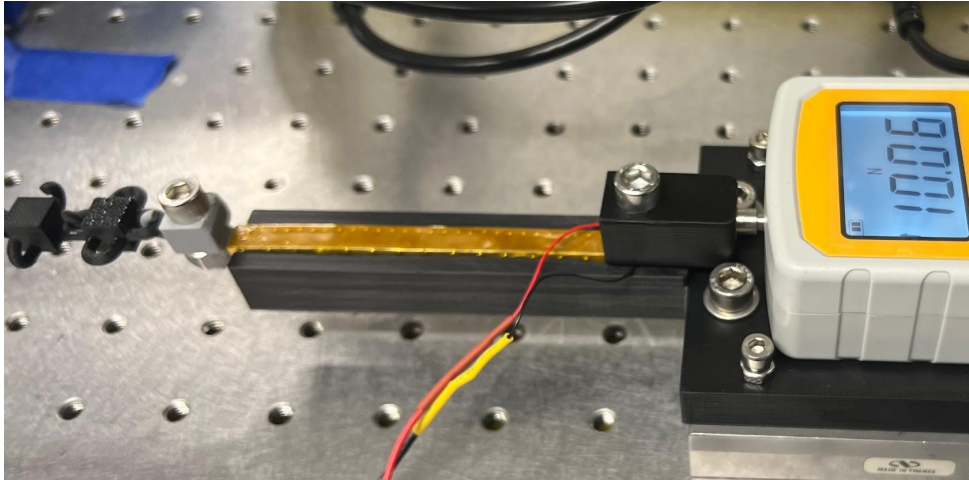
PAM Unit Cell: Materials



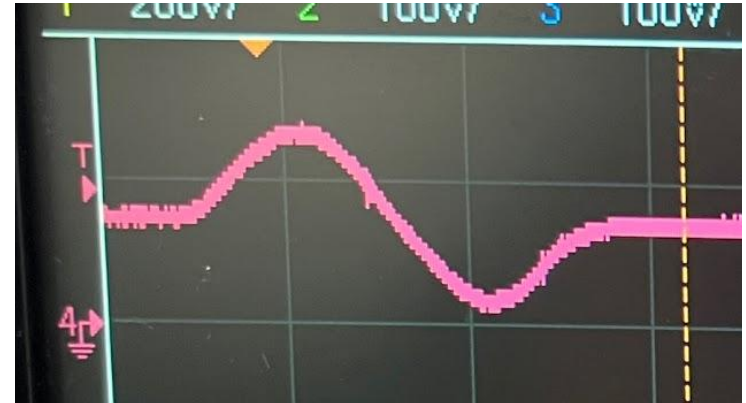
Material	Total Price	Lead Time	Yield Strength (MPa)
PLA (printed on campus)	Few cents - few dollars	2-6 hours	30-60
Nylon	\$37.49	6-10 days	40-70
Aluminum	\$79.75	6-10 days	~240
SS	\$62.78	6-10 days	40-95
Polycarbonate	\$34.06	7-13 days	~60
Glass filled nylon	\$54.85	15 days	230-300

Piezoelectric Actuator

- Tens of micrometers level actuation and control
- Input different waveforms of pulses



Actuator Voltage

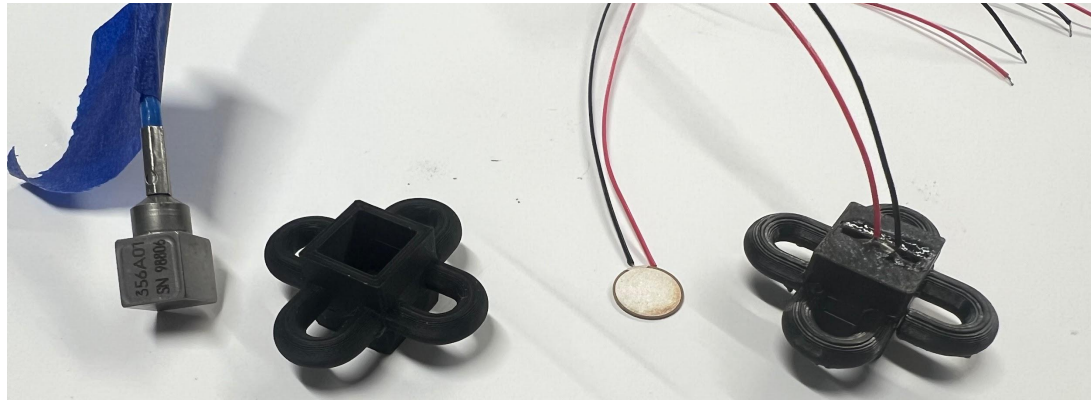


0ms 0.5 ms 1.0 ms 1.5 ms

Time (us)

Accelerometer vs. Ceramic Piezo Force Sensor

	Triaxial Accelerometer	Piezo Force Sensor
Measured Quantity	Acceleration	Quickly Changing Compression or Tension Force
Dimensions of Measurement	3	1
Calibration	Precalibrated	Needs Calibration



Accelerometer vs. Ceramic Piezo Force Sensor

Triaxial Accelerometer:

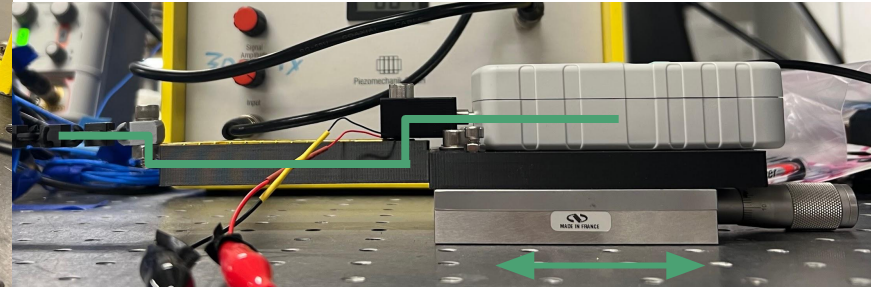
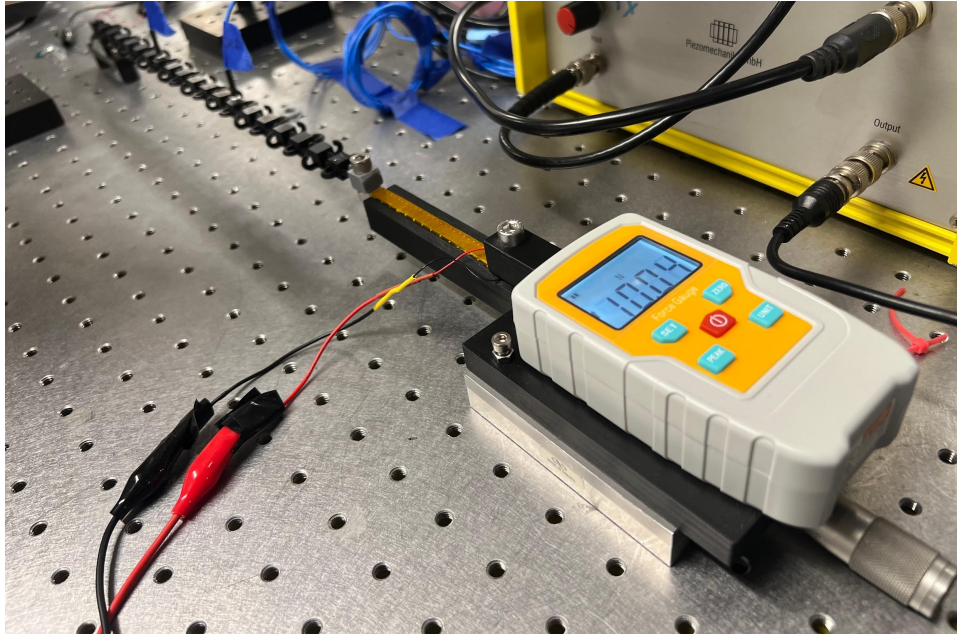
- Measures acceleration in each dimension
- 3 dimensions
- Useful for other modes of wave pulses
- Better for 2D case (eventually)
- Precalibrated

Piezo Force Sensor:

- Approximately measures compressional or tensile force
- But constant offset issue is present, may be inconsistent
- Calibration curve needed

	Triaxial Accelerometer	Piezo Force Sensor
Measured Quantity	Acceleration	Quickly Changing Compression or Tension Force
Dimensions of Measurement	3	1

Tension Sensor + Linear Stage

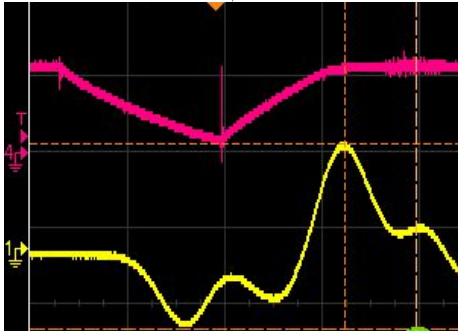


Consequences of Imperfect Impulse

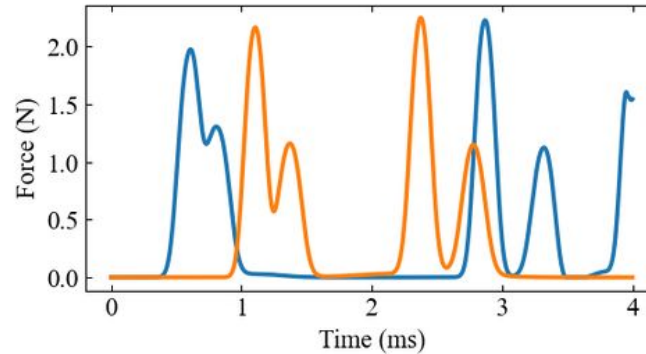
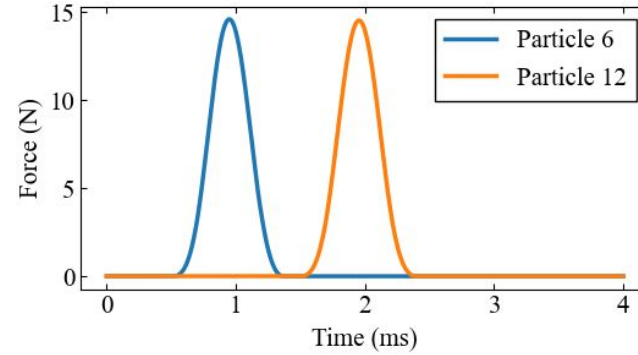
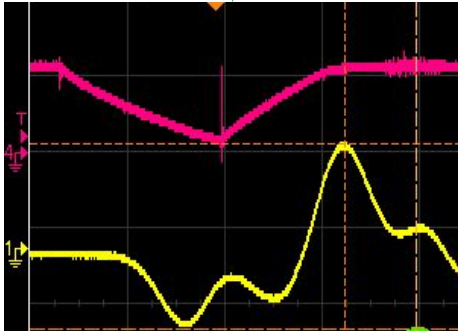
Input
Voltage
(Desired)



Actuator
Voltage

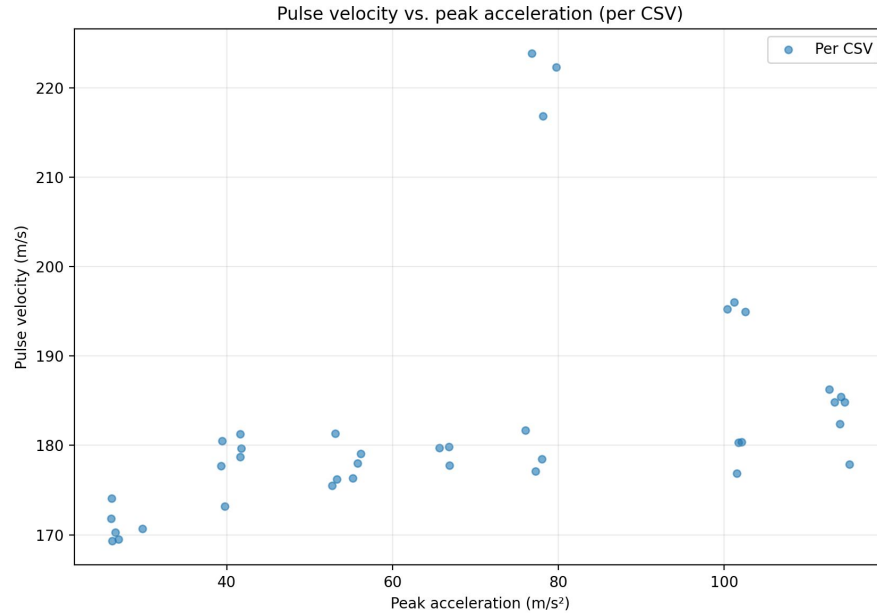


Pulse
Readings
from Sensor

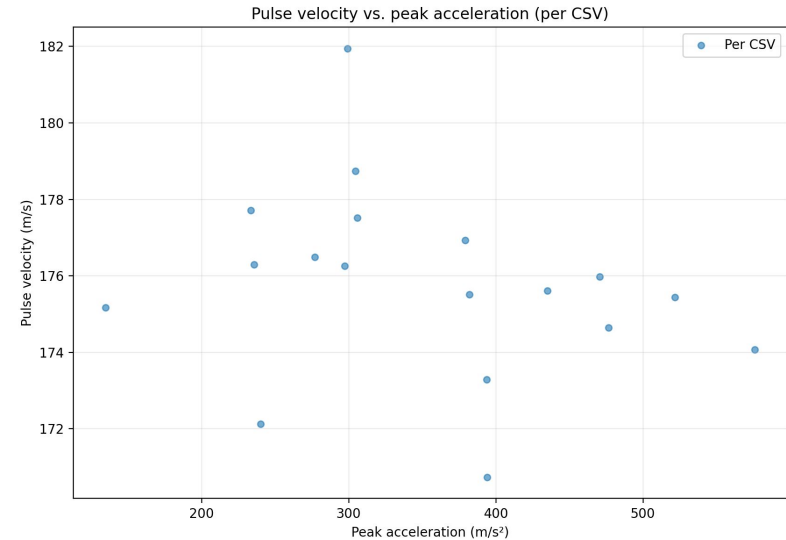


Velocity vs. Amplitude Experiment

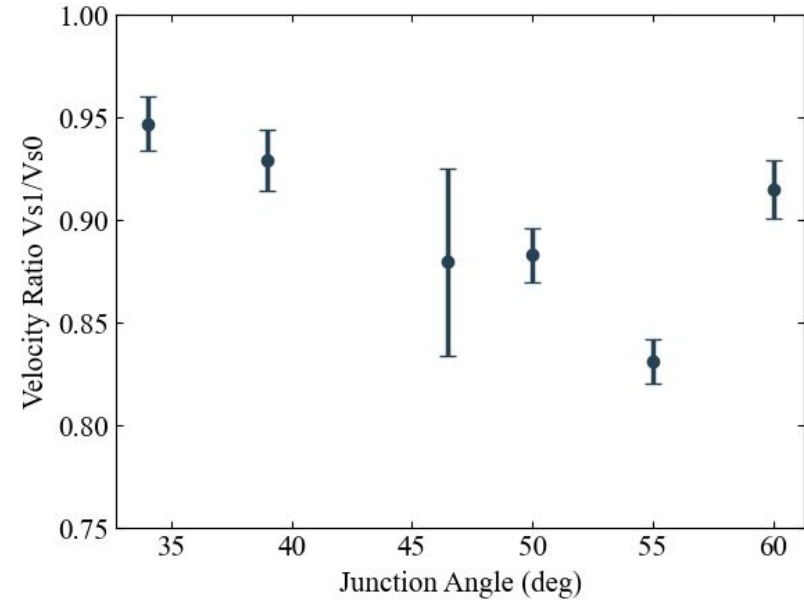
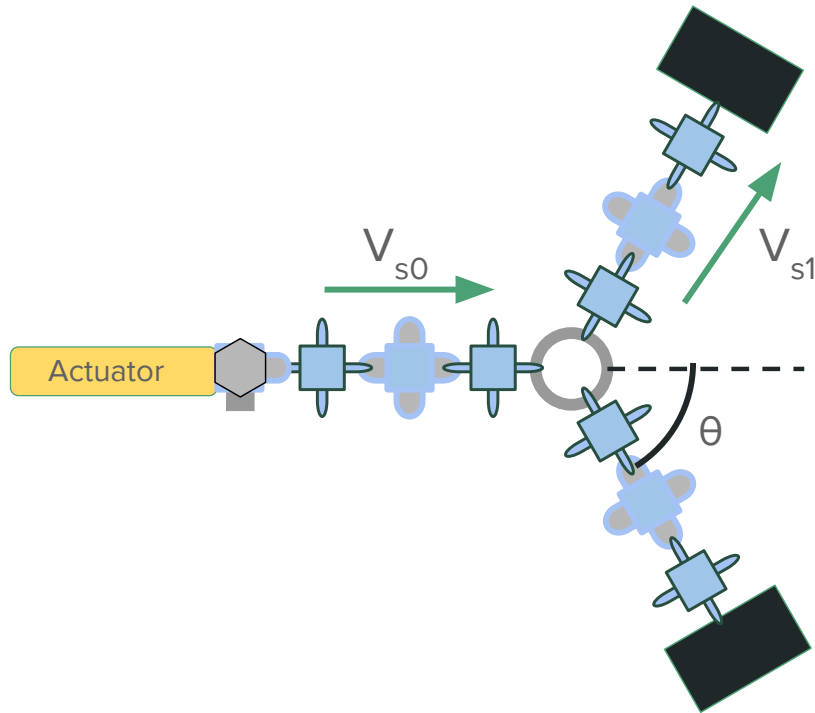
actuator



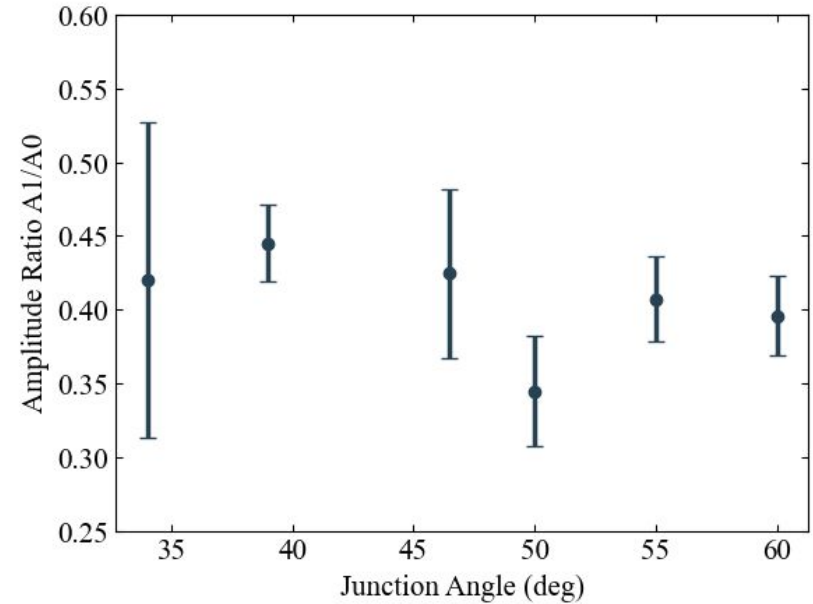
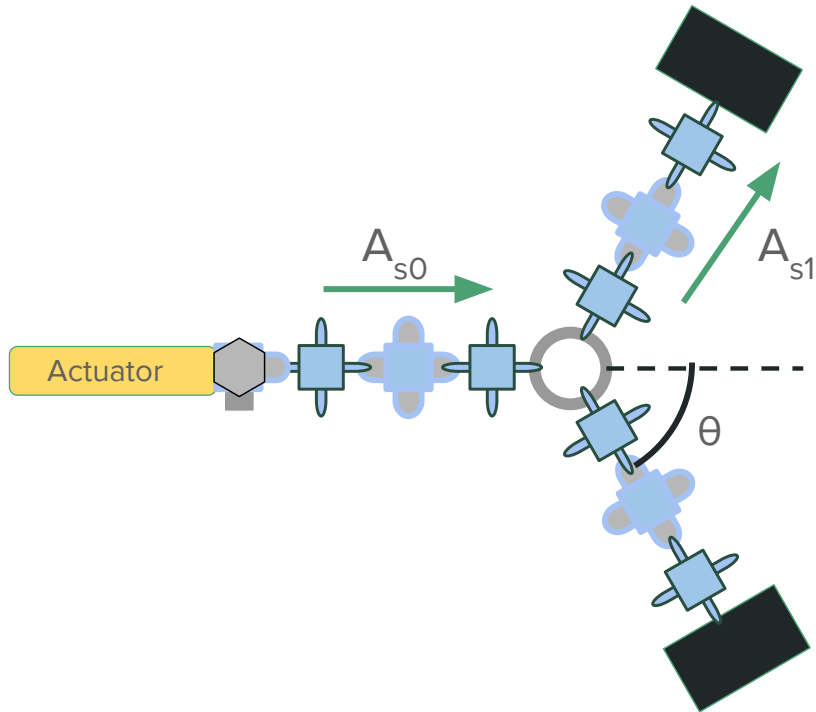
manual



Junction with Symmetrical Angles – Velocity Comparison



Junction with Symmetrical Angles – Amplitude Comparison



Interesting Sine Wave Response with Aluminum

